CSE 270 Sports Analytics

Pythagorean wins and winning percentage

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Pythagorean wins and winning percentage

Libraries used

```
library(ggplot2)
library(dplyr)
library(SportsAnalytics270)
```

Major League of Baseball



Major League of Baseball: American League

American League

Division	Team	City
East	Baltimore Orioles	Baltimore, MD
	Boston Red Sox	Boston, MA
	New York Yankees	New York City, NY
	Tampa Bay Rays	St. Petersburg, FL
	Toronto Blue Jays	Toronto, ON
Central	Chicago White Sox	Chicago, IL
	Cleveland Indians	Cleveland, OH
	Detroit Tigers	Detroit, MI
	Kansas City Royals	Kansas City, MO
	Minnesota Twins	Minneapolis, MN
West	Houston Astros	Houston, TX
	Los Angeles Angels	Anaheim, CA
	Oakland Athletics	Oakland, CA
	Seattle Mariners	Seattle, WA
	Texas Rangers	Arlington, TX

Major League of Baseball

Regular Season

The Major League Baseball season runs from the start of April to the end of September, with each club playing 162 games.

Teams normally play "series" of three (occasionally four) games on consecutive days against the same opponent, normally with a "homestand" of two or three series, or a road trip.

Playoffs

At the end of the regular season the three divisional winners and the wild-card team (from each league go forward into the playoffs. The four teams in each league play two series of knockout baseball to determine the who.

The two league winners then meet in the "World Series" (so called, because the first sponsor was the "Daily World" newspaper).

Major League of Baseball

World Series

- Is played between the winners of the playoffs of the American League and the National League.
- Is a best-of-seven tournament.
- World Series is won by the team who will win 4 games out of 7.

Baseball: The game

Baseball rules explained

https://www.youtube.com/playlist?list=PLAj29Nf-IKG7V5z9zg3SSJmPVccUtNhQX

https://www.youtube.com/watch?v=E160D9rEY0M

Baseball: Father of Sabermetrics

- Sabermetrics is the empirical analysis of baseball, especially baseball statistics that measure in-game activity.
- Sabermetrics aims to quantify baseball players' performances based on objective statistical measurements.
- The term is derived from the acronym SABR, which stands for the Society for American Baseball Research.

More about Sabermetrics: https://sabr.org/sabermetrics
https://www.mlb.com/news/sabermetrics-in-baseball-a-casual-fans-guide

Baseball: Moneyball

Anyone knows these guys ?





Baseball: Moneyball

What about this 2 guys ?



Baseball: Moneyball

- Moneyball(2011) is an American movie based on the true story of the Oakland Athletics baseball team and their general manager, Billy Beane.
- Billy Beane(starred by Brad Pitt) is a former professional baseball player. He is well known for his innovative approach to player evaluation during his time as the general manager of the Oakland Athletics team.
- Billy Beane is one of the pioneers of Sabermetrics.
- Peter Brand(starred byJonah Hill) is a fictional character in the book and film Moneyball. He is based on Paul DePodesta, a real-life baseball executive who worked with Billy Beane at the Oakland Athletics.
- DePodesta, like Brand, was a Harvard graduate with a background in economics. He joined the Athletics in 2000 and became the team's assistant general manager. DePodesta played a key role in developing and implementing the sabermetric approach that the team used to build a competitive roster on a limited budget.

Baseball Data

- Lahman season-by-season dataset
  - From authors website
    http://www.seanlahman.com/baseball-archive/statistics/
- Retrosheets game-by-game data, play-by-play data http://www.retrosheet.org/
- PITCHf/x pitch-by-pitch data

Runs and winning percentage

Runs and winning percentage

Lahman teams dataset

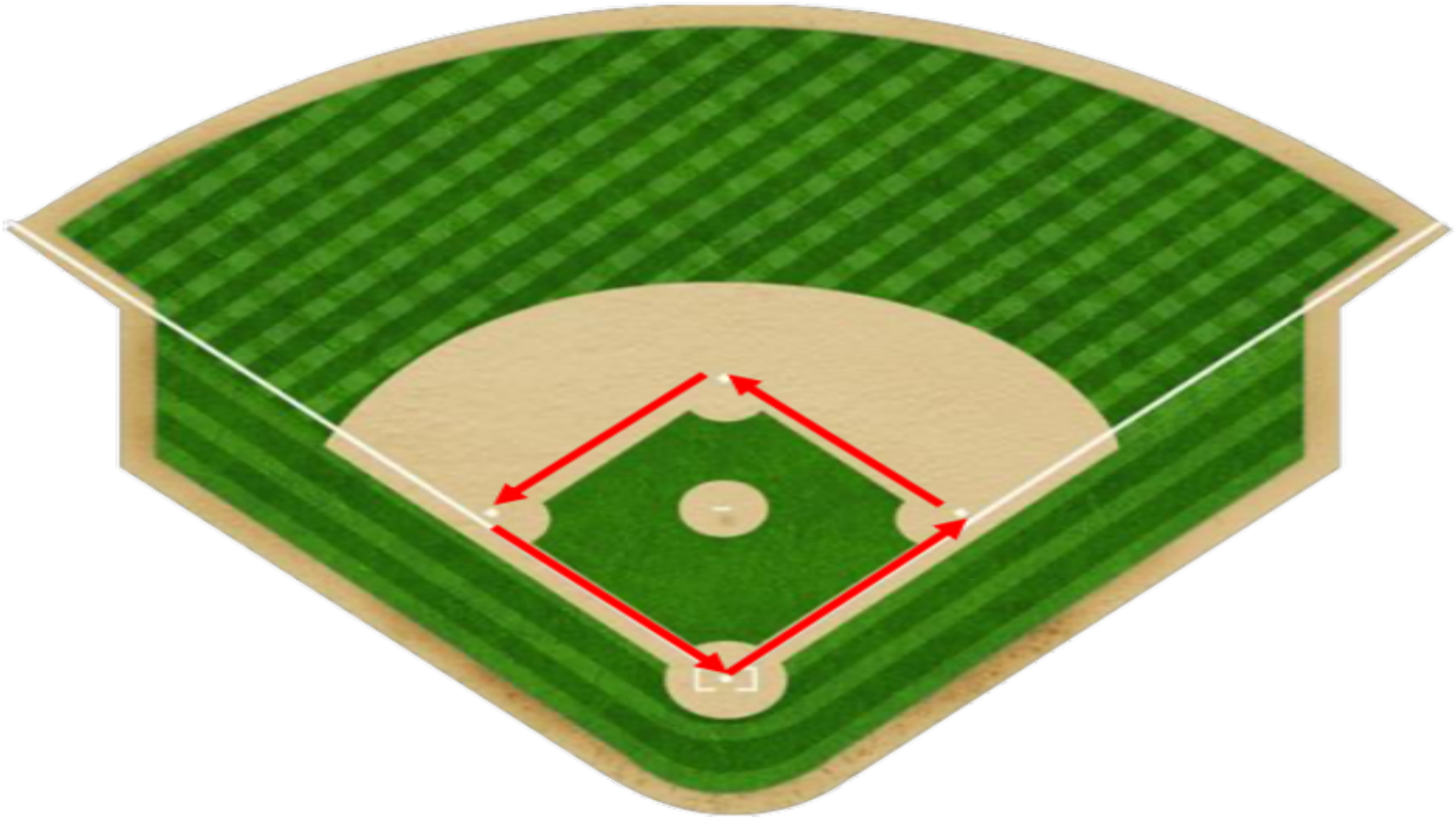
```
library(SportsAnalytics270)
head(mlb_standings)
```

```
##   Season Lig Div           Team  W  L   G   R  RA Rank
## 1   1999  AL   W Anaheim Angels 70 92 162 711 826    4
## 2   2000  AL   W Anaheim Angels 82 80 162 864 869    3
## 3   2001  AL   W Anaheim Angels 75 87 162 691 730    3
## 4   2002  AL   W Anaheim Angels 99 63 162 851 644    2
## 5   2003  AL   W Anaheim Angels 77 85 162 736 743    3
## 6   2004  AL   W Anaheim Angels 92 70 162 836 734    1
```

Pythagorean wins

In baseball, a run is scored when a player advances around first, second and third base and returns safely to home plate, touching the bases in that order, before three outs are recorded and all obligations to reach base safely on batted balls are met or assured.

The team that scores more runs, wins



Pythagorean wins

We are going to model the winning percentage based on Runs allowed and Runs created.

- Create two variables:
  Runs differential: Runs-Runs Allowed
  Winning percentage

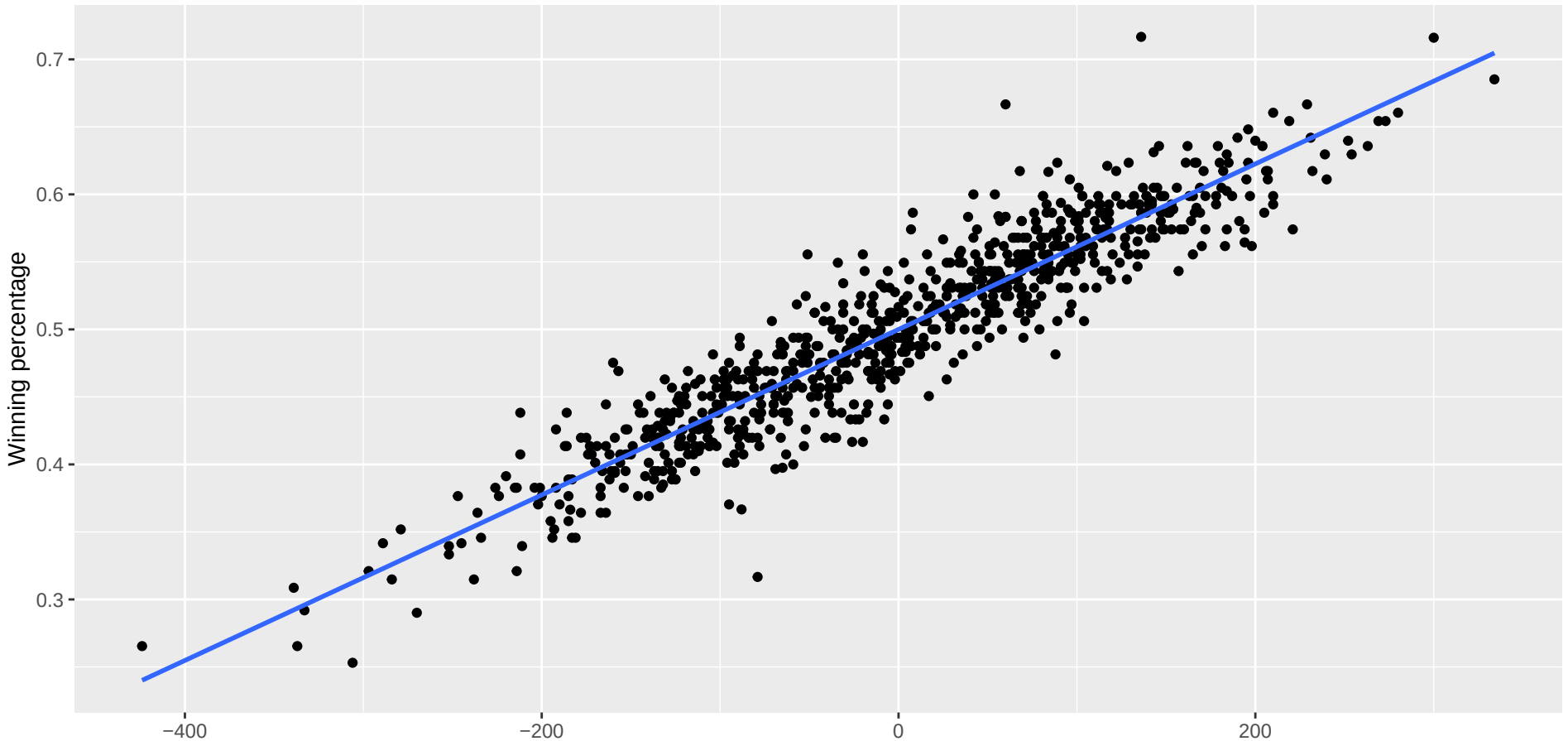
Create Winning Percentage and Run differential

```
# Winning percentage
mlb_standings$Wpct<-mlb_standings$W/(mlb_standings$W+mlb_standings$L)
# Run differential
mlb_standings$RD<-mlb_standings$R-mlb_standings$RA
```

Pythagorean wins

The scatterplot shows clear relationship between Run differential and Winning Percentage.

```
library(ggplot2)
ggplot(mlb_standings, aes(x=RD, y=Wpct))+geom_point()+
  geom_smooth(method = "lm", se=F)+
  labs(x="Run differential",y="Winning percentage")
```



Pythagorean wins

A simple linear regression model with Winning percentage as a dependent and Runs differential as an independent variable.

$$Wnpct = \beta_0 + \beta_1 * RD + \epsilon$$

```
options(scipen=999)
mod<-lm(Wpct~RD, data=mlb_standings)
summary(mod)
```

```
##
## Call:
## lm(formula = Wpct ~ RD, data = mlb_standings)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.134894 -0.017365  0.000152  0.016872  0.133322
##
## Coefficients:
##                Estimate  Std. Error t value            Pr(>|t|)
## (Intercept) 0.499983953 0.000960385  520.61 <0.0000000000000002 ***
## RD          0.000612948 0.000008411   72.87 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02733 on 808 degrees of freedom
## Multiple R-squared:  0.8679, Adjusted R-squared:  0.8678
## F-statistic:  5310 on 1 and 808 DF,  p-value: < 0.00000000000000022
```

Pythagorean wins

How will you interpret intercept and slope ?

```
coefficients(mod)
```

```
##  (Intercept)           RD
## 0.4999839529 0.0006129485
```

Pythagorean wins

```
coefficients(mod)
```

```
##  (Intercept)           RD
## 0.4999839529 0.0006129485
```

How will you interpret intercept and slope ?

Intercept: Intercept: If the run differential is 0, thus the team makes and allows an equal number of runs, then you expect to win 50% of the games.

slope: 1 unit increase in run differential increases the winning percentage by 0.000623.

Pythagorean wins

The predicted value for the Winning percentage can be out of Range [0:1]

```
coefficients(mod)
```

```
##  (Intercept)           RD
## 0.4999839529 0.0006129485
```

What will be your prediction for a team with a Run differential of 850?

Pythagorean formula for Winning percentage

It is derived empirically by Bill James, the founder of Sabermetrics

$$Wpct = \frac{R^k}{R^k + RA^k}$$

R - runs made
RA - runs allowed

The number of the wins calculated by Pythagorean formula is called ***Pythagorean wins.***

Generic Pythagorean formula

$$Wpct = \frac{R^k}{R^k + RA^k}$$

The equation can be rewritten in the following way

$\frac{W}{L} = \frac{R^k}{RA^k}$ $log(\frac{W}{L}) = k * log(\frac{R}{RA})$

Now we can fit a regression model to find the best value for k

According to Wayne Winston's estimates, the best value for k is 1.9

The logic behind Pythagorean formula

- The team wins games in proportion of their "quality" to win.
- Lets say the quality of the team to win is defined by the Run ration

$$\frac{\text{Runs}}{\text{Runs Allowed}}$$

- Team A with runs of 60 and runs allowed of 40 will have quality ratio of 60/40
- Then quality measure of the opponent (all teams collectively) will be 40/60
- Thus to calculate the probability of the team to win:

$$\frac{60/40}{\frac{60}{40}+\frac{40}{60}} = \frac{60^2}{60^2+40^2}$$

Wayne Winston modification

$$Wpct = \frac{R^k}{R^k + RA^k}$$

$$Wpct = \frac{Rt^k}{Rt^k + 1} \text{ where, Rt is the runs ratio}$$

$$Rt = \frac{\text{Runs}}{\text{Runs allowed}}$$

Usage

$$Wpct = \frac{Rt^k}{Rt^k + 1}$$

Suppose a team has scored 850 runs and allowed 800 runs last season.

Rt=850/800=1.0625

Using Pythagorean formula, the predicted winning percentage will be 0.53 or 85.9 games (162*0.53)

Suppose the team trades player A who has created 150 runs for a player who has created 170.

All other things equal this will result the team to score 870 runs. What will happen with winning percentage ?

Rt=870/800 and the predicted winning percentage 0.5418 which translates to 87.7 games.

Buying the player will help the team, to win 2 more games.

Calculate some accuracy measure

Root Mean Square error

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (\hat{y}_i - y_i)^2}{n}}$$

```
mlb_standings$PW_percentage<-mlb_standings$R^1.9/
  (mlb_standings$R^1.9+mlb_standings$RA^1.9)

errors<-mlb_standings$Wpct-mlb_standings$PW_percentage

sqrt(mean(errors^2))
```

```
## [1] 0.02547051
```

Our average error is 2.5%, or 0.025

Or it will predict wrongly 4 games (162*0.025)

Pythagorean wins

Now compare error from Pythagorean calculations vs error from regression model

```
mod<-lm(Wpct~RD, data=mlb_standings)
# mod$ residuals gives you the error
sqrt(mean(mod$residuals^2))
```

```
## [1] 0.02729928
```

Pythagorean wins

Bad or good luck ?

https://www.baseball-reference.com/leagues/majors/2025-standings.shtml

```
mlb_standings$PW <- mlb_standings$PW_percentage*(mlb_standings$W+mlb_standings$L)
mlb_standings$PW <- round(mlb_standings$PW,0)
```

```
mlb_standings %>%
  filter(Season == 2025)%>%
  select(Team, W, PW ) %>%
  head()
```

```
##                    Team  W PW
## 1 Arizona Diamondbacks 80 82
## 2             Athletics 76 73
## 3       Atlanta Braves 76 80
## 4    Baltimore Orioles 75 69
## 5       Boston Red Sox 89 93
## 6    Chicago White Sox 60 71
```

Pythagorean wins for other sports

Pythagorean wins for other sports

Pythagorean wins in NBA

```
library(SportsAnalytics270)
data(nba_east)
data(nba_west)
nba_st <- rbind(nba_east,nba_west)
```

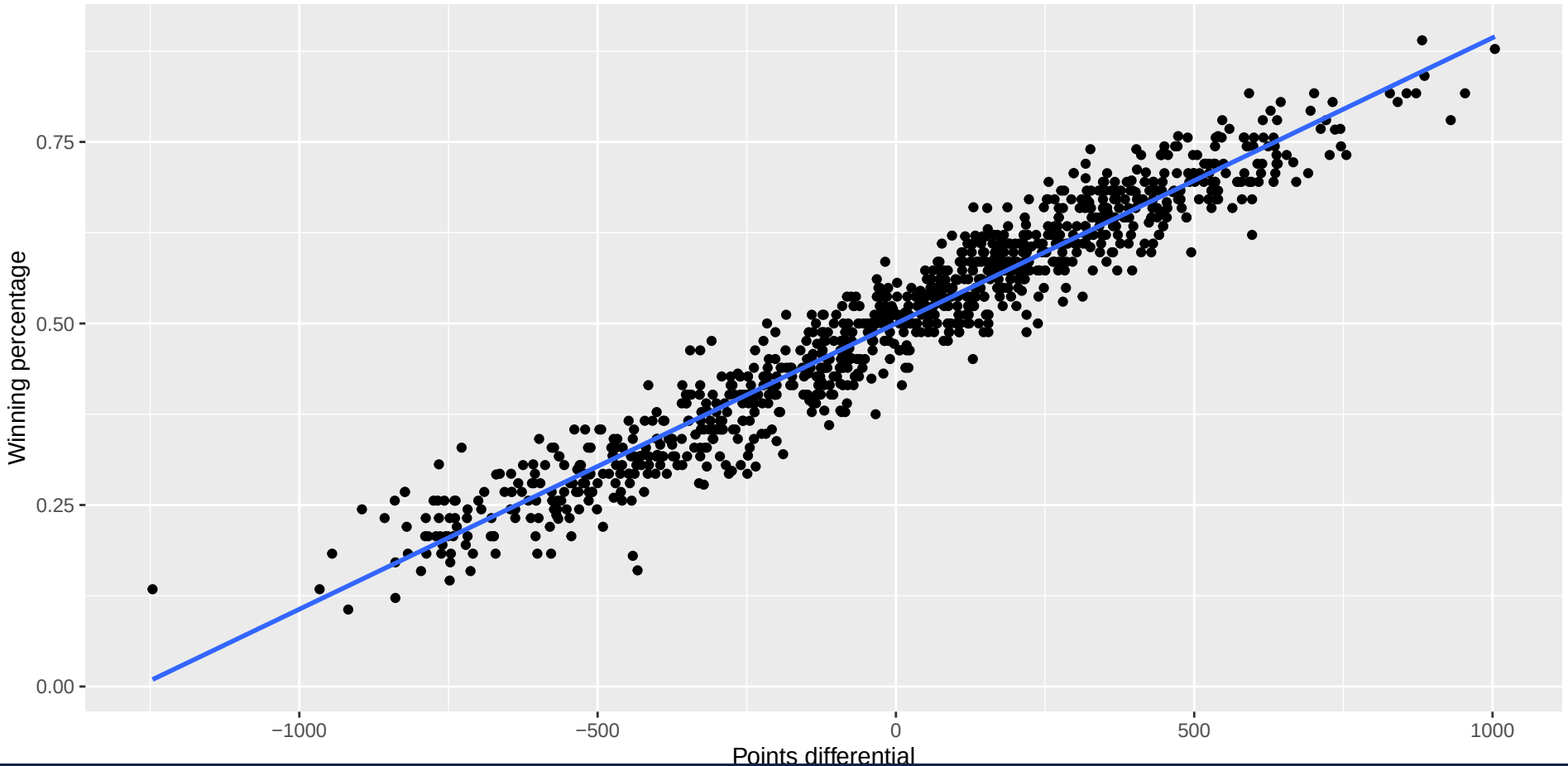
```
head(nba_st)
```

```
##   Rank                Team  W  L   Pct   GB Season   PS   PA
## 1    1       Chicago Bulls 61 21 0.744 <NA>   1991 9024 8278
## 2    2      Boston Celtics 56 26 0.683    5   1991 9145 8668
## 3    3     Detroit Pistons 50 32 0.610   11   1991 8205 7937
## 4    4     Milwaukee Bucks 48 34 0.585   13   1991 8727 8524
## 5    5 Philadelphia 76ers 44 38 0.537   17   1991 8641 8656
## 6    6       Atlanta Hawks 43 39 0.524   18   1991 9003 8940
```

Pythagorean wins in NBA

```
nba_st$PD <- nba_st$PS - nba_st$PA
```

```
ggplot(nba_st, aes(x=PD, y=Pct))+geom_point()+
  geom_smooth(method = "lm", se=F)+
  labs(x="Points differential",y="Winning percentage")
```



Pythagorean wins in NBA

```
options(scipen=999)
model2 <- lm(Pct~PD, data=nba_st)
summary(model2)
```

```
##
## Call:
## lm(formula = Pct ~ PD, data = nba_st)
##
## Residuals:
##       Min         1Q     Median         3Q        Max
## -0.169570 -0.025864  0.000637  0.026347  0.124351
##
## Coefficients:
##                Estimate  Std. Error t value             Pr(>|t|)
## (Intercept) 0.499958091 0.001253901   398.7 <0.0000000000000002 ***
## PD          0.000393506 0.000003341   117.8 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04016 on 1024 degrees of freedom
## Multiple R-squared:  0.9313, Adjusted R-squared:  0.9312
## F-statistic: 1.387e+04 on 1 and 1024 DF,  p-value: < 0.00000000000000022
```

Generic Pythagorean formula

Recall:

$$Wpct = \frac{R^k}{R^k + RA^k}$$

$$\frac{W}{W+L} = \frac{R^k}{R^k+RA^k} => \frac{W+L}{W} = \frac{R^k+RA^k}{R^k} => \frac{W}{W} + \frac{L}{W} = \frac{R^k}{R^k} + \frac{RA^k}{R^k} => 1 + \frac{L}{W} = 1 + \frac{RA^k}{R^k} =>$$
$$\frac{L}{W} = \frac{RA^k}{R^k} = (\frac{RA}{R})^k => \frac{W}{L} = (\frac{R}{RA})^k => log(\frac{W}{L}) = k * log(\frac{R}{RA})$$

- We want to find the value of k
- General linear regression equation is: $y = \beta_0 + \beta_1 X$, however here we have no intercept model, thus $y = \beta_1 X$

Pythagorean wins in NBA

0 at the beginning of the right hand of the regression equation, indicates that we are fitting no intercept model

```
nba_st$Ratio <- nba_st$PS/nba_st$PA
```

```
model2 <- lm(log(W/L)~0+log(Ratio), data=nba_st)
```

Pythagorean wins in NBA

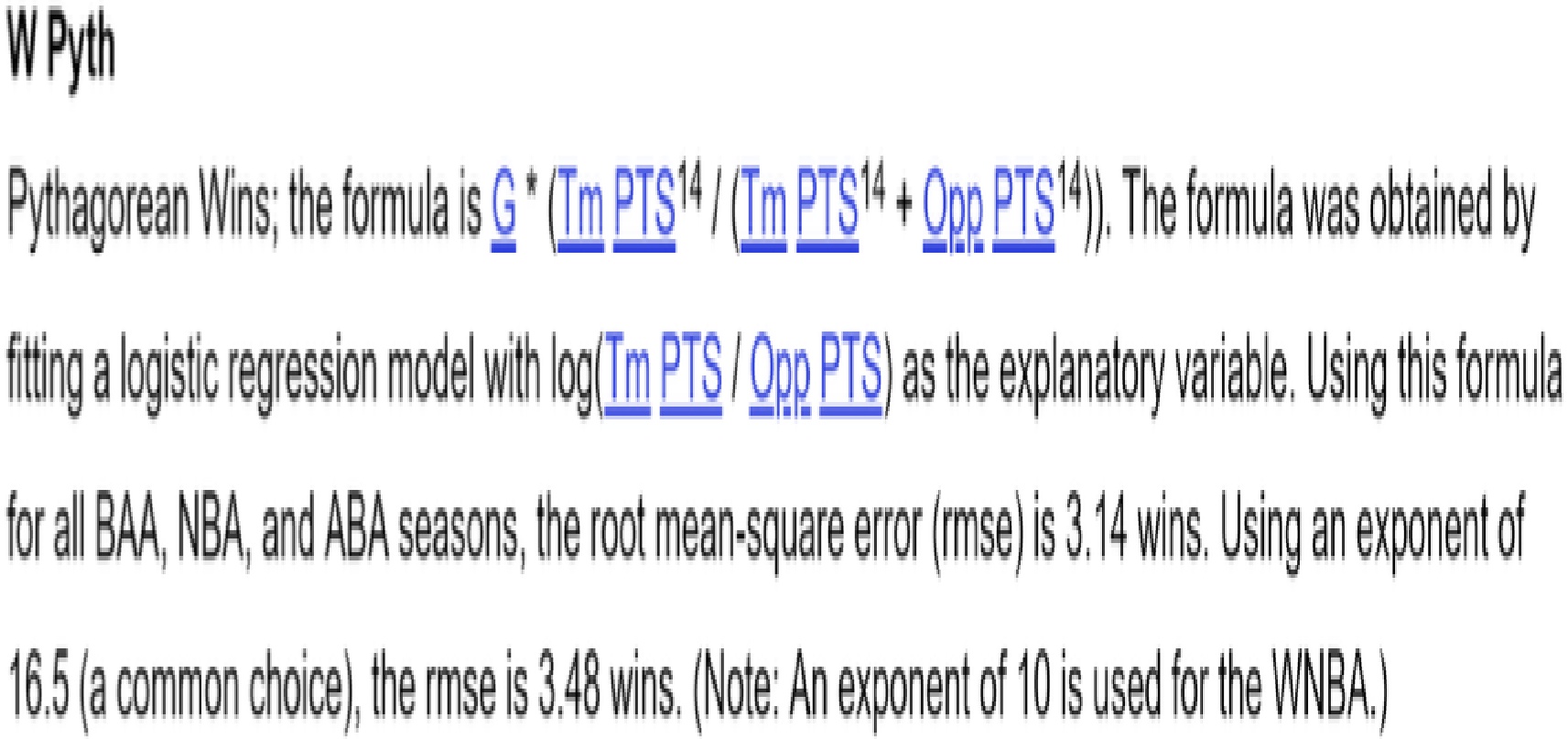
```
summary(model2)
```

```
##
## Call:
## lm(formula = log(W/L) ~ 0 + log(Ratio), data = nba_st)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57133 -0.11675 -0.00137  0.10222  0.71264
##
## Coefficients:
##             Estimate Std. Error t value            Pr(>|t|)
## log(Ratio)    14.045      0.114   123.2 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1682 on 1025 degrees of freedom
## Multiple R-squared:  0.9368, Adjusted R-squared:  0.9367
## F-statistic: 1.519e+04 on 1 and 1025 DF,  p-value: < 0.00000000000000022
```

Pythagorean wins in NBA

The estimated value of K is 14.093

```
k = 14.093
```



from: https://www.basketball-reference.com/about/glossary.html#wins_pyth

Pythagorean wins in NBA

```
nba_st$PW <- nba_st$PS^k/(nba_st$PS^k+nba_st$PA^k)
sqrt(mean((nba_st$PW-nba_st$Pct)^2))
```

```
## [1] 0.03710989
```

```
82*0.0367104
```

```
## [1] 3.010253
```

On average we are going to be "wrong" by 3 games per team

Pythagorean wins in NBA

Lets have a look at the last season only

Create variable PW_G, Pythagorean expected wins

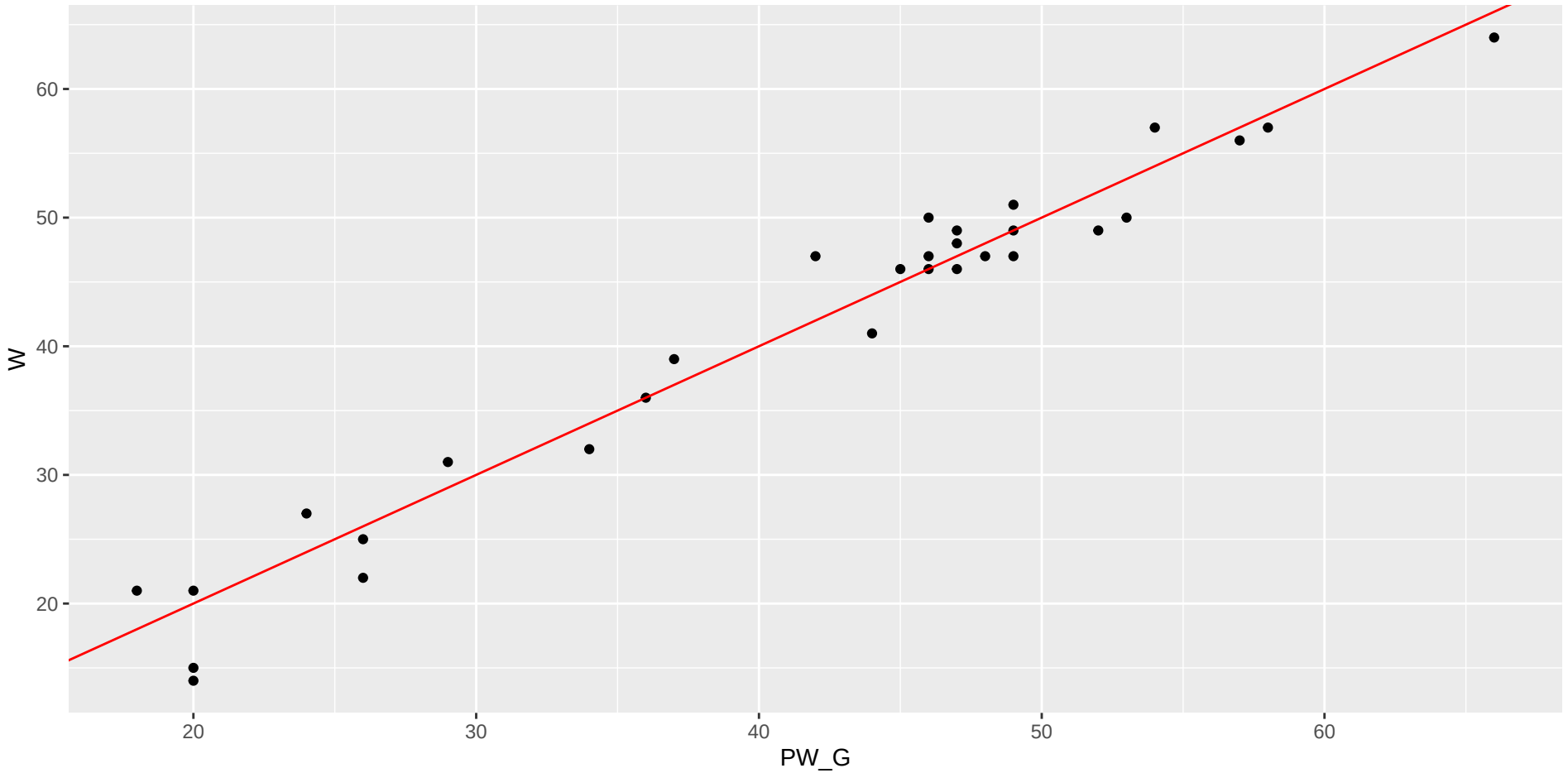
```
nba_st$G <- nba_st$W+nba_st$L
nba_st$PW_G <- round(nba_st$PW*nba_st$G)
nba_st %>% filter(Season==2024) %>% head()
```

```
##   Rank                Team  W  L   Pct   GB Season    PS   PA  PD    Ratio
## 1    1      Boston Celtics 64 18 0.780    -   2024  9887 8957 930 1.103829
## 2    2     New York Knicks 50 32 0.610 14.0   2024  9249 8873 376 1.042376
## 3    3     Milwaukee Bucks 49 33 0.598 15.0   2024  9756 9541 215 1.022534
## 4    4 Cleveland Cavaliers 48 34 0.585 16.0   2024  9236 9038 198 1.021908
## 5    5       Orlando Magic 47 35 0.573 17.0   2024  9058 8892 166 1.018668
## 6    6      Indiana Pacers 47 35 0.573 17.0   2024 10110 9860 250 1.025355
##           PW  G PW_G
## 1 0.8009405 82   66
## 2 0.6421928 82   53
## 3 0.5778737 82   47
## 4 0.5757643 82   47
## 5 0.5648008 82   46
## 6 0.5873140 82   48
```

Pythagorean wins in NBA

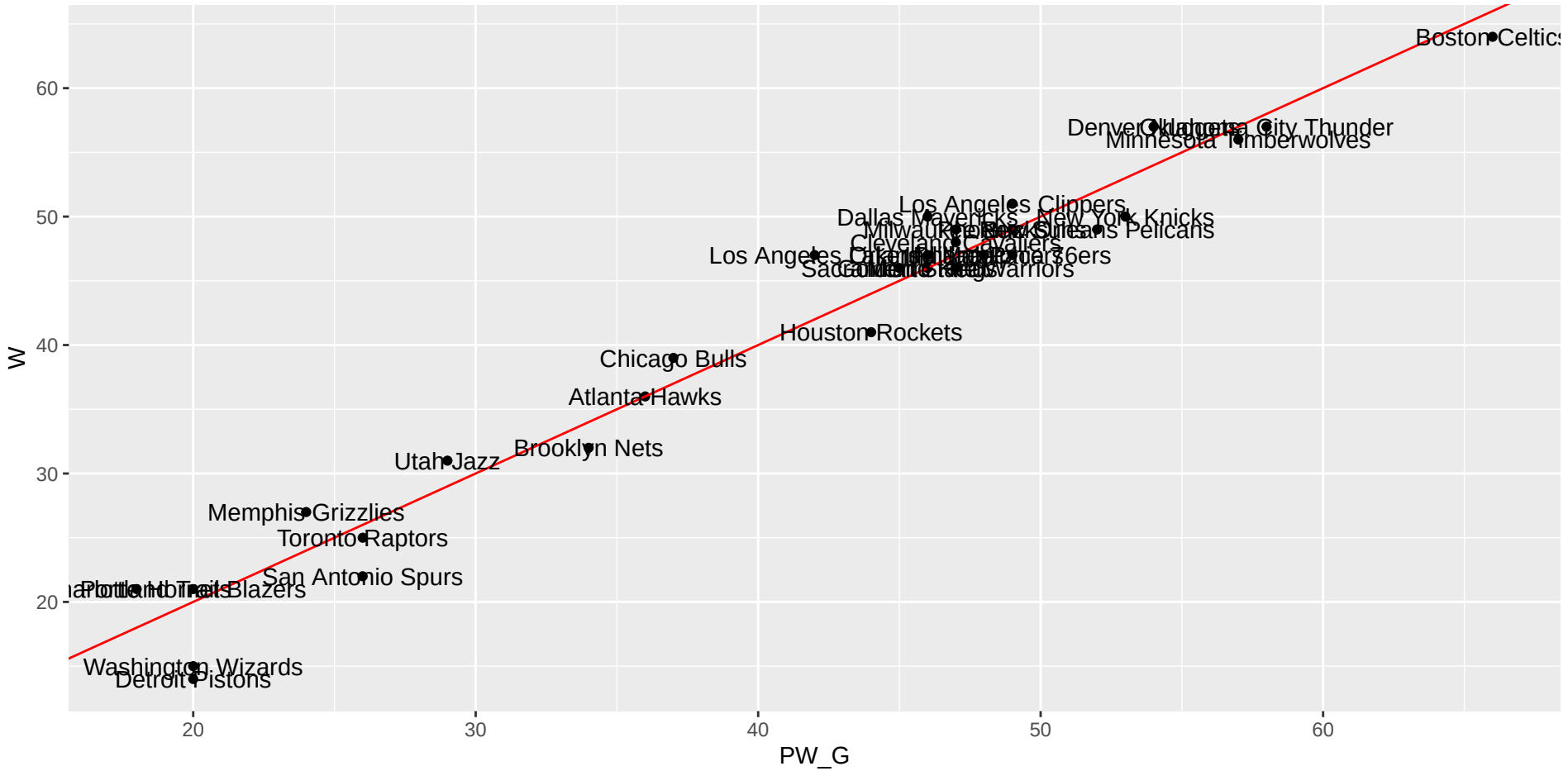
teams above the line are “lucky”, below the line are “unlucky”

```
nba_st %>% filter(Season==2024)%>%
  ggplot(aes(x=PW_G, y=W))+geom_point()+
  geom_abline(intercept =0 , slope = 1, col='red')
```



Pythagorean wins in NBA

```
nba_st %>% filter(Season==2024)%>%
  ggplot(aes(x=PW_G, y=W, label=Team))+geom_point()+
  geom_abline(intercept =0 , slope = 1, col='red')+
  geom_text()
```



Pythagorean wins for other sports

Pythagorean wins are also calculated for other sports.

in NFL

$$Win = \frac{\text{points for}^{2.37}}{\text{points for}^{2.37} + \text{points against}^{2.37}}$$